RAVEN DESIGN DRAFT — COMMENTS AND DRAFT RESPONSES

These are proposed responses. Items marked “Need more info — TBD” require additional information or team decisions.

1. PRIOR ART

Instructor comment:
“Multiple examples are presented, but in single clauses with no description of how it works or the tradeoffs.”

Response:
Expand the examples to explain how their architectures work and their tradeoffs. VTA combines a reusable GEMM core, tensor ALU, DMA, and local memory, requiring careful scheduling and buffering. Gemmini uses a RISC-V-controlled systolic array and banked scratchpad, with configurable dataflows and tight processor integration. FINN generates network-specific streaming hardware, exploiting layer parallelism while tailoring the implementation to the mapped network. Explain how these approaches inform RAVEN’s shared compute architecture.

Sources:
VTA: https://tvm.apache.org/2018/07/12/vta-release-announcement.html
Gemmini: https://github.com/ucb-bar/gemmini
FINN: https://xilinx.github.io/finn/

2. DIFFERENTIATION

Instructor comment:
“You describe this as "a lasting education tool", but have nothing to really support that. For maintenance, add a handoff for after graduation ,pinned tool versions, etc.”

Response:
Support the educational-tool claim with concrete deliverables: documented RTL, testbenches, reproducible build scripts, example workloads, and tested tool versions. Explain how another student could reproduce and extend the reference design.

Need more info — TBD:
Handoff recipient, post-graduation maintenance owner, and exact pinned tool versions.

3. TANGIBLE IMPACT

Instructor comment:
“First three positive impacts are generic read as risks. You have no supporting information to substantiate your claims. Your claims are also heavily hedged ("potential", "may").”

Response:
Replace unsupported economic and environmental claims with verifiable project outcomes: local inference without a cloud inference service, a reusable FPGA implementation, and measured architectural tradeoffs. Claims about reduced latency, energy, or cost should identify a comparison baseline and supporting evidence.

Need more info — TBD:
Comparison baselines and measurements supporting performance, energy, or cost claims.

4. ETHICAL CONSIDERATIONS

Instructor comment:
“None? Bias and privacy of software models does not constitute ethical considerations of the hardware you are proposing and the potential for misuse and privacy problems.”

Response:
Discuss how programmable inference hardware could enable surveillance or continuous analysis of sensitive inputs. Address memory-access risks from programmable DMA and explain that local processing alone does not guarantee privacy. Describe the controls the team plans to implement and how demonstrations will handle potentially sensitive data.

Need more info — TBD:
Live versus prerecorded inputs, recording and storage policies, access restrictions, and planned hardware/software protections.

5. DRAWBACKS AND LIMITATIONS

Instructor comment:
“I would like to see severity/likeliness of each.”

Response:
Add severity, likelihood, mitigation, and an observable warning condition for each listed risk.

Need more info — TBD:
Team-reviewed ratings for resource limits, bandwidth bottlenecks, timing closure, unsupported operators, quantization loss, and integration complexity.

6. PERFORMANCE EXPECTATIONS AND REQUIREMENTS

Instructor comment:
“No real targets for array size,GOPs, inferences/sec, or accuracy?”

Response:
The proposed starting target is an 8×8 array containing 64 INT8 MAC units, with architectural INT32 accumulation and a clock frequency of at least 100 MHz. Assuming one MAC per unit per cycle, the theoretical arithmetic ceiling is 6.4 GMAC/s, or 12.8 GOPS when multiplication and

addition count separately. These are proposed targets and analytical ceilings, not measured results.

Need more info — TBD:
Team approval of the proposed targets, selected models and input sizes, inference-rate requirements, and accuracy thresholds.

7. HARDWARE AND SOFTWARE DEPENDENCIES

Instructor comment:
“Essentially no dependencies listed (hardware + software).”

Response:
List the hardware and software required to build, program, test, and run RAVEN. Proposed dependencies include the Ultra96-V2, power and programming/debug connections, Vivado, an RTL simulator, the selected RISC-V core and compatible toolchain, model-processing/reference libraries, and any selected AXI/DMA IP.

Need more info — TBD:
Exact RISC-V core, DMA implementation, simulator, libraries, and tested versions.

8. TARGET DATES

Instructor comment:
“No calendar dates?”

Response:
Replace semester-relative descriptions with specific calendar dates.

Need more info — TBD:
Dates for subsystem completion, first integration, model demonstrations, beta, and final submission/demo.

9. INDIVIDUAL ACCOUNTABILITY

Instructor comment:
“Who is doing what? We need individual accountability and owners ("entire team" does not suffice).”

Response:
State individual subsystem ownership explicitly:

Michael: Compute architecture, MAC datapath, pipeline, compute RTL, and compute verification.

Quan: Supporting operators, requantization, pooling, Element-wise Add, and operator verification.

Taylor: BRAM organization, memory interfaces, DMA, and memory verification.

Nahjay: RISC-V integration, MMIO, control, and runtime.

Jackie: Model analysis/export, tensor extraction, and golden-reference generation.

Need more info — TBD:
One accountable lead for each shared integration milestone and the final handoff.

10. ACCESSIBILITY, USABILITY, AND MAINTENANCE

Instructor comment:
“We need a handoff plan or documentation. Just saying it will be amazing and last an eternity is not a plan.”

Response:
Provide a tagged release, tested dependency versions, setup/build/programming instructions, interface documentation, known-input tests with expected outputs, troubleshooting guidance, and documented limitations. A proposed handoff acceptance check is having someone other than the subsystem author reproduce a known test using the documentation.

Need more info — TBD:
Handoff recipient, maintenance owner, and agreement on the handoff process.

11. TASK, FEATURE, AND MILESTONE DESCRIPTIONS

Instructor comment:
“We need acceptance criteria.”

Response:
Give each milestone a measurable pass condition. Proposed examples include:

- Compute outputs match designated integer golden references across supported test cases.
- DMA transfers reproduce the expected bytes for supported lengths and boundaries.
- RISC-V launches an operation and receives correct completion status and results.
- Three selected models pass their defined functional and quality checks.
- The integrated FPGA design meets its agreed clock and timing requirements.

Need more info — TBD:
Model-specific quality thresholds and final milestone dates.

12. MOCKUP CONTENT

Instructor comment:
“Register map has no DMA registers, PARAM_DIM packing is undefined, Element-wise Add has no opcode, and there’s no software-side or dataflow mockup.”

Response:
Document the DMA control interface alongside the NPU register interface. Define dimension fields and packing, include Element-wise Add in the operation map, and add software execution-flow and tensor-dataflow diagrams. Clearly distinguish proposed interface definitions from finalized ones.

Need more info — TBD:
Selected DMA implementation, register addresses, field widths and packing, opcode values, and agreed completion behavior.

13. MOCKUP READABILITY

Instructor comment:
“Small text, pixelated (come on, use vector graphics). Long and narrow (I have to search up and down looking for things).”

Response:
Replace screenshots with SVG or vector-PDF diagrams. Separate the system overview, compute-core detail, and software/dataflow views. Use readable labels at the final printed size, consistent block styles, and clearly labeled interfaces. Check every figure at normal viewing size before submission.